

Maps

$$\rightarrow \phi^i : \underset{\substack{\uparrow \\ \xi}}{\Sigma} \rightarrow \underset{\substack{\uparrow \\ \phi_i}}{\mathcal{S}} \quad \text{target space} \quad \xi^\mu = (\xi^{++}, \xi^-)$$

$$\rightarrow \mathcal{S} = \int \underset{\Sigma}{d^2 \xi} \partial_\mu \phi^i \partial^\mu \phi^j \left(\underset{\text{sym}}{g_{ij}(\phi)} \eta^{\mu\nu} + \underset{\text{a-sym}}{b_{ij}(\phi)} \epsilon^{\mu\nu} \right) \quad \begin{matrix} \text{metric} \\ \text{B-field} \sim \text{specific to 2d} \end{matrix}$$

$$\rightarrow \delta \phi^\mu :$$

$$0 = -2g_{\mu k} \left\{ \left[\partial_\mu \partial_\nu \phi^k + \frac{1}{2} g^{kl} (2g_{l(i,j)} - g_{ij,l}) \partial_\mu \phi^i \partial_\nu \phi^j \right] \eta^{\mu\nu} + \frac{1}{2} g^{kl} \mathcal{H}_{ij} \partial_\mu \phi^i \partial_\nu \phi^j \epsilon^{\mu\nu} \right\}$$

$\mathcal{H}_{ij} = b_{li,j} + b_{ij,l} + b_{jl,i}$
totally a-sym.

$$= -2g_{\mu k} \nabla_{++}^{(+)} \partial_\mu \phi^k \quad \nabla_{++}^{(+)} \rightarrow \nabla^{(0)} + \Gamma$$

$$\rightarrow \mathcal{S} \sim \int \underset{\Sigma}{d^2 \xi} \partial \phi^i g_{ij} \partial \phi^j + \int \underset{V}{d^3 \xi} \mathcal{H}_{ijk} \partial_a \phi^i \partial_b \phi^j \partial_c \phi^k \epsilon^{abc}$$

$\partial V = \Sigma$ $a, b, c = (+, -, 3)$

$$\rightarrow (g_{ij}, \mathcal{H}_{ijk}, \Gamma = \Gamma^{(0)} + \Gamma)$$

$\rightarrow \mathcal{S}$ carries torsionful Riemann geometry

Susy $\{Q, Q\} \sim P$ (transl.), $[P, M] \sim P$ (Lorentz), $[Q, M] = Q$
 $[Q, P] \sim 0$

$$\rightarrow (2,2) \text{ susy in 2d: } \begin{cases} \{Q_+, \bar{Q}_+\} = 2i \partial_{++} \\ \{Q_-, \bar{Q}_-\} = 2i \partial_{--} \end{cases}$$

$$\rightarrow (\xi^{++}, \xi^{--}, \theta^+, \bar{\theta}^+, \theta^-, \bar{\theta}^-), \quad \{\theta, \theta\} = 0$$

→ superfields $\phi(\xi, \theta, \bar{\theta})$

~ spinorial covariant derivatives

→ differential operators $(\mathcal{D}_{\pm}, \bar{\mathcal{D}}_{\pm})$ s.t.:

$$\{\mathcal{D}, Q\} = 0, \quad \{\mathcal{D}_+, \bar{\mathcal{D}}_+\} = 2i \partial_{++}, \quad \{\mathcal{D}_-, \bar{\mathcal{D}}_-\} = 2i \partial_{--}$$

→ constrain ϕ 's : $\bar{\mathcal{D}}_{\pm} \phi = 0$ (chiral field)

$$\bar{\mathcal{D}}_+ \chi = \mathcal{D}_- \chi = 0 \quad (\text{twisted chiral field})$$

define generalized
Kähler
geometry

$$\begin{cases} \bar{\mathcal{D}}_+ \bar{\mathcal{D}}_- \Sigma_{\phi} = 0 & (\text{complex linear field}) \\ \bar{\mathcal{D}}_+ \mathcal{D}_- \Sigma_{\chi} = 0 & (\text{complex twisted linear field}) \end{cases}$$

$$\mathcal{D}_+ l = 0$$

$$\mathcal{D}_- \kappa = 0$$

} left / right semi-chiral fields

can write $\Sigma_{\phi} = l + \kappa, \Sigma_{\chi} = l + \bar{\kappa}$

→ components : usually $\phi = \varphi(\xi) + \theta^+ \psi_+(\xi) + \dots$ ↑

cumbersome, unless formalism up to some $\theta^+ \bar{\theta}^+ \theta^- \bar{\theta}^-$ terms

$$\Rightarrow \text{set } \theta = 0 \Rightarrow \phi|_{\theta=0} = \varphi(\xi)$$

$$\mathcal{D}_{\pm} \phi|_{\theta=0} = \psi_{\pm}(\xi) \quad \sim \text{complex-valued}$$

$$\mathcal{D}_+ \mathcal{D}_- \phi|_{\theta=0} = \mathcal{F}(\xi) \sim \text{auxiliary field}$$

$$\rightarrow i \mathcal{D}_{\pm}|_{\theta=0} = Q_{\pm}|_{\theta=0} = i \frac{\partial}{\partial \theta^{\pm}} = i \int d\theta^{\pm}$$

$$Q_{\pm} = i \frac{\partial}{\partial \theta^{\pm}} + \frac{1}{2} \theta^{\pm} \mathcal{D}_{\pm\pm} \sim \partial / \partial \xi^{\pm\pm}$$

$$\rightarrow \delta \phi = \epsilon^+ Q_+ \phi + \dots$$

$$\begin{aligned} \Rightarrow \delta \varphi(\xi) &= \delta \phi|_{\theta=0} = \epsilon^+ Q_+ \phi|_{\theta=0} = i \epsilon^+ \mathcal{D}_+ \phi|_{\theta=0} \\ &= i \epsilon^+ \psi_+(\xi) \end{aligned}$$

→ take a chiral $\phi, \bar{\phi}$

$$\Rightarrow S = \int d^2 \bar{\epsilon} d\theta^+ d\bar{\theta}^+ d\theta^- d\bar{\theta}^- K(\phi^i, \bar{\phi}^{\bar{j}})$$

$$= \int d^2 \bar{\epsilon} \frac{\partial}{\partial \theta^+} \frac{\partial}{\partial \bar{\theta}^+} \frac{\partial}{\partial \theta^-} \frac{\partial}{\partial \bar{\theta}^-} K(\phi^i, \bar{\phi}^{\bar{j}})$$

$$= \int d^2 \bar{\epsilon} \mathcal{D}_+ \bar{\mathcal{D}}_+ \mathcal{D}_- \bar{\mathcal{D}}_- K(\phi^i, \bar{\phi}^{\bar{j}})$$

$$\sim \int d^2 \bar{\epsilon} \underbrace{K_{, \phi^i \bar{\phi}^{\bar{j}}}}_{\text{metric}} \partial_{++} \phi^i \partial_{++} \bar{\phi}^{\bar{j}} \quad \text{sigma model complex fields } \phi$$

→ this is Kähler geometry, with Kähler potential K
an example of complex geometry of the target space

Complex geometry (M, J, g) endomorphism, real

$$\rightarrow J: T(M) \rightarrow T(M), \quad J^2 = -1$$

almost complex structure $\Rightarrow \dim M$ is even

$$\rightarrow \text{define projection operators } \pi_{\pm} = \frac{1}{2}(1 \pm iJ)$$

$$\Rightarrow \text{Lie bracket } \pi_{\mp} [\pi_{\pm} u, \pi_{\pm} v] = 0$$

$u, v \in T(\text{target space})$

$$\Rightarrow \underbrace{J_m^l J_{[n, l]}^i - J_n^l J_{[m, l]}^i}_{\equiv N_{mn}^i} = 0$$

$\equiv N_{mn}^i$ Nijenhuis tensor

$$\rightarrow N_{mn}^i = 0 \Leftrightarrow J \text{ integrable}$$

$$\rightarrow g \text{ is Hermitian w.r.t. } J \text{ iff } J^T g J = g$$

$$\Rightarrow J^i_j g_{ik} J^k_l = g_{jl}$$

$$\underbrace{J^i_j g_{ik}}_{= \omega_{jk}} = -g_{jl} J^l_k$$

$$\Rightarrow \omega_{jk} = -\omega_{kj}$$

$$\rightarrow \omega = \frac{1}{2} \omega_{li} d\phi^i \wedge d\phi^l = \omega_{\bar{a}a} d\phi^a \wedge d\bar{\phi}^{\bar{a}}$$

write in complex coord.

canonical coord.

complex

$$\Rightarrow J = \begin{pmatrix} i\delta^a_b & 0 \\ 0 & -i\delta^{\bar{a}}_{\bar{b}} \end{pmatrix}, \quad g_{\text{new}} \Rightarrow J = \begin{pmatrix} 0 & g_{a\bar{b}} \\ g_{\bar{a}b} & 0 \end{pmatrix}$$

$\rightarrow \omega$ symplectic in Kähler geometry, $d\omega = 0$

$$\Rightarrow \omega = -i g_{a\bar{b},z} d\phi^a \wedge d\bar{\phi}^{\bar{b}} \wedge d\phi^z$$

$$+ i g_{\bar{a}b,\bar{z}} d\bar{\phi}^{\bar{a}} \wedge d\phi^b \wedge d\phi^{\bar{z}}$$

$$\rightarrow g_{\bar{b}[a,z]} = 0, \quad g_{\bar{b}a} = \partial_a \bar{\gamma}_{\bar{b}}$$

$$\Rightarrow g_{b[a,\bar{z}]} = 0 = \partial_{\bar{b}} \bar{\gamma}_{[a,\bar{z}]} \Rightarrow \bar{\gamma} = \partial_{\bar{a}} k$$

$$g_{a\bar{b}} = \partial_a \partial_{\bar{b}} (K + \underbrace{f(\phi) + \bar{f}(\bar{\phi})}_{\text{some hol. \& a-hol. func.}})$$

Kähler potential

some hol. & a-hol. func.

$\rightarrow$ in the (2,2) sigma model, the chiral superfields are canonical coord.

$$\Rightarrow J = \begin{pmatrix} i\delta & 0 \\ 0 & -i\delta \end{pmatrix}, \quad K(\phi, \bar{\phi}) = \mathcal{L}(\text{lagrangian})$$

• Add twisted chiral fields $\bar{\mathcal{D}}_+ \chi = 0 = \mathcal{D}_- \chi$

$$\begin{aligned} \rightarrow \mathcal{I} &\sim \int d^2 \bar{\epsilon} d^4 \theta \mathcal{K}(\phi^a, \bar{\phi}^{\bar{a}}, \chi^{a'}, \bar{\chi}^{\bar{a}'}) \\ &= \int d^2 \bar{\epsilon} d^4 \theta \Big|_{\theta=0} \mathcal{K}(\phi^a, \bar{\phi}^{\bar{a}}, \chi^{a'}, \bar{\chi}^{\bar{a}'}) \\ &= \int d^2 \bar{\epsilon} \left(\mathcal{K}_{, a \bar{a}} \partial_{(++)} \phi^a \partial_{(=-)} \bar{\phi}^{\bar{a}} - \mathcal{K}_{, a' \bar{a}'} \partial_{(++)} \chi^{a'} \partial_{(=-)} \bar{\chi}^{\bar{a}'} \right. \\ &\quad \left. + \mathcal{K}_{, a a'} \partial_{[++] \phi^a \partial_{(=-)} \chi^{a'} - \mathcal{K}_{, \bar{a} \bar{a}'} \partial_{[++] \bar{\phi}^{\bar{a}} \partial_{(=-)} \bar{\chi}^{\bar{a}'} + \dots \right) \end{aligned}$$

$$\rightarrow \mathcal{K}^{(3)} = \mathcal{K}_{a b' \bar{c}} d\phi^a \wedge d\chi^{b'} \wedge d\bar{\chi}^{\bar{c}} + \dots \quad (\text{physically relevant object})$$

$$\rightarrow \text{metric} \quad g_{A\bar{B}} = \left(\begin{array}{c|c} 0 & \mathcal{K}_{a\bar{b}} \\ \hline \mathcal{K}_{\bar{a}b} & 0 \end{array} \right)$$

$$\begin{aligned} \rightarrow \text{complex structures:} \quad \mathcal{I}_{(+)} &= \left(\begin{array}{c|c} i & 0 \\ \hline 0 & -i \end{array} \right) \\ \mathcal{I}_{(-)} &= \left(\begin{array}{c|c} -i & 0 \\ \hline 0 & i \end{array} \right) \Rightarrow \mathcal{I}_{(\pm)}^\dagger g \mathcal{I}_{(\pm)} = g \end{aligned}$$

• Bi-Hermitian geometry : $[\mathcal{I}_{(+)}, \mathcal{I}_{(-)}] = 0$

$$\rightarrow \mathcal{P} = \mathcal{I}_{(+)} \mathcal{I}_{(-)} \quad (\text{projection}) \quad \mathcal{P}^2 = \mathcal{P} \quad \text{local prod. structure}$$

$$\Rightarrow \omega_{(\pm)} = g \mathcal{I}_{(\pm)} \quad , \quad d\omega_{(\pm)} \neq 0 \quad (\text{not K\"ahler})$$

generalised K\"ahler

• Add all the other fields $\Rightarrow$ get generalised K\"ahler

chiral, twisted chiral, complex linear, complex twisted linear,
left / right semi-chiral